

SAVONLINNA, Finland

WMO: 029480

Lat: 61.946N

Lon: 28.932E

Elev: 313

StdP: 14.53

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-11.2	-23.4	1.5	-17.8	-16.0	2.3	-11.3	17.8	23.0	16.1	24.3	1.8	270	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	80.5	65.7	77.0	63.4	73.6	61.7	67.7	75.9	65.6	73.4	63.7	71.1	8.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	93.0	70.7	62.7	86.1	69.0	60.7	80.2	66.9	32.2	76.4	30.6	73.1	29.2	71.7	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.6	13.3	DB	-22.9	83.8	7.0	4.3	-27.9	86.9	-32.0	89.4	-35.9	91.8	-41.0	94.9	(4)
(5)			WB	-23.1	70.2	6.9	2.7	-28.1	72.1	-32.2	73.6	-36.1	75.1	-41.1	77.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	16.8	17.4	25.0	37.2	48.6	56.6	62.0	59.0	50.4	39.5	30.6	23.7		
	DBStd	17.96	13.59	13.04	9.51	6.54	8.00	6.72	5.91	5.82	6.01	6.95	8.18	11.65		
	HDD50	5041	1028	914	776	387	123	16	1	5	67	329	581	814		
	HDD65	9538	1493	1334	1241	833	508	264	129	199	437	790	1031	1279		
	CDD50	1042	0	0	0	4	81	215	374	284	80	4	0	0		
	CDD65	64	0	0	0	0	1	12	37	14	0	0	0	0		
	CDH74	626	0	0	0	0	40	134	332	120	0	0	0	0		
	CDH80	102	0	0	0	0	2	20	67	13	0	0	0	0		
	WSAvg	6.3	6.2	6.1	6.5	6.1	6.5	6.5	5.8	5.5	6.0	6.4	7.0	7.2		
	PrecAvg	24.7	1.9	1.5	1.3	1.1	1.7	2.6	2.8	2.8	2.2	2.4	2.2	2.0		
(15) Precipitation	PrecMax	30.2	3.0	2.9	2.6	2.9	5.2	4.3	5.7	5.2	5.6	5.3	4.5	3.4		
	PrecMin	18.3	0.7	0.2	0.3	0.2	0.4	0.9	0.8	0.7	0.7	0.6	0.4	0.9		
	PrecStd	2.7	0.7	0.7	0.6	0.6	1.1	0.8	1.2	1.3	1.2	1.1	0.9	0.7		
	PrecStd	2.7	0.7	0.7	0.6	0.6	1.1	0.8	1.2	1.3	1.2	1.1	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	39.6	49.3	64.4	79.0	82.6	85.6	82.3	69.4	56.2	48.3	44.5		
		MCWB	36.5	36.5	39.9	50.9	59.7	64.6	68.8	67.2	60.3	52.8	46.6	42.9		
	2%	DB	35.9	36.4	42.7	58.0	73.6	78.0	81.7	77.7	65.5	53.3	45.4	40.6		
		MCWB	35.0	34.9	36.8	47.6	57.2	62.8	67.1	64.4	58.6	51.0	44.2	39.4		
	5%	DB	34.1	34.5	39.0	53.4	68.6	73.7	77.8	74.3	62.7	51.6	43.3	37.8		
		MCWB	33.4	33.6	34.8	43.8	54.8	60.6	64.8	62.9	57.1	49.6	42.3	36.7		
	10%	DB	32.5	33.2	36.3	49.2	64.2	69.9	74.1	70.8	60.2	49.8	41.2	35.8		
		MCWB	32.0	32.6	33.4	41.8	52.2	58.5	62.9	61.8	55.9	48.1	40.3	35.1		
	0.4%	WB	36.9	37.0	40.8	53.8	63.0	68.1	71.4	69.9	62.1	53.8	46.9	42.6		
		MCDB	37.2	38.3	47.2	60.8	74.1	78.0	81.1	78.4	65.5	55.5	47.8	43.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.1	35.3	37.4	49.2	59.5	65.0	68.7	66.4	60.0	51.8	44.3	39.3		
		MCDB	35.6	36.3	41.3	55.8	68.7	74.4	77.9	73.6	63.8	53.0	45.0	40.0		
	5%	WB	33.6	33.8	35.5	45.2	56.5	62.4	66.7	64.7	58.2	50.1	42.3	36.8		
		MCDB	34.0	34.3	37.7	51.7	66.6	70.5	74.7	72.1	61.7	51.2	43.1	37.6		
	10%	WB	32.1	32.5	34.0	42.1	53.6	60.1	64.6	62.8	56.4	48.1	40.3	35.1		
		MCDB	32.6	33.0	35.7	47.8	62.3	67.9	71.7	69.3	59.5	49.2	41.1	35.8		
	5% DB	MDBR	10.4	11.7	14.2	16.0	18.9	17.2	16.7	15.9	12.8	8.7	7.0	8.4		
		MCDBR	7.0	8.5	15.8	24.1	26.4	23.5	22.4	22.0	17.2	9.9	7.4	6.5		
		MCWBR	7.0	7.7	11.1	14.0	12.7	10.6	9.4	10.4	11.1	7.8	7.3	6.3		
		MCWBR	7.1	7.7	12.6	21.5	21.9	20.1	19.6	18.8	14.9	8.9	7.4	6.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	7.1	7.3	9.6	13.3	11.8	10.0	9.5	9.9	10.3	7.7	7.3	6.1		
		MCWBR	7.1	7.3	9.6	13.3	11.8	10.0	9.5	9.9	10.3	7.7	7.3	6.1		
	Clear-Sky Solar Irradiance	taub	0.243	0.291	0.318	0.358	0.354	0.350	0.375	0.359	0.336	0.317	0.261	N/A		
		taud	2.034	2.117	2.195	2.340	2.446	2.490	2.435	2.461	2.514	2.403	2.122	N/A		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	136	207	248	263	275	278	267	261	247	204	128	N/A		
		Edh at Noon	12	22	30	32	31	30	31	28	23	17	11	N/A		
(44) All-Sky Solar Radiation	RadStd	RadAvg	70	248	719	1158	1562	1682	1620	1243	705	278	74	34		
		RadStd	11	61	102	133	183	100	153	148	81	41	12	4		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	-322	N/A	N/A	N/A

Nomenclature: See separate page